

Day 1: Tasks

1.

List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.



Scenarios	How the multimeter helps
Computer not Powering on	Measures the power supply voltage to check if the PSU is providing the correct output.
Smartphone not charging	Checks the charger, USB cable, or battery voltage to find where the charging problem exists.
Laptop Battery issue	Measures battery voltage to determine whether the battery is charged, discharged, or faulty.

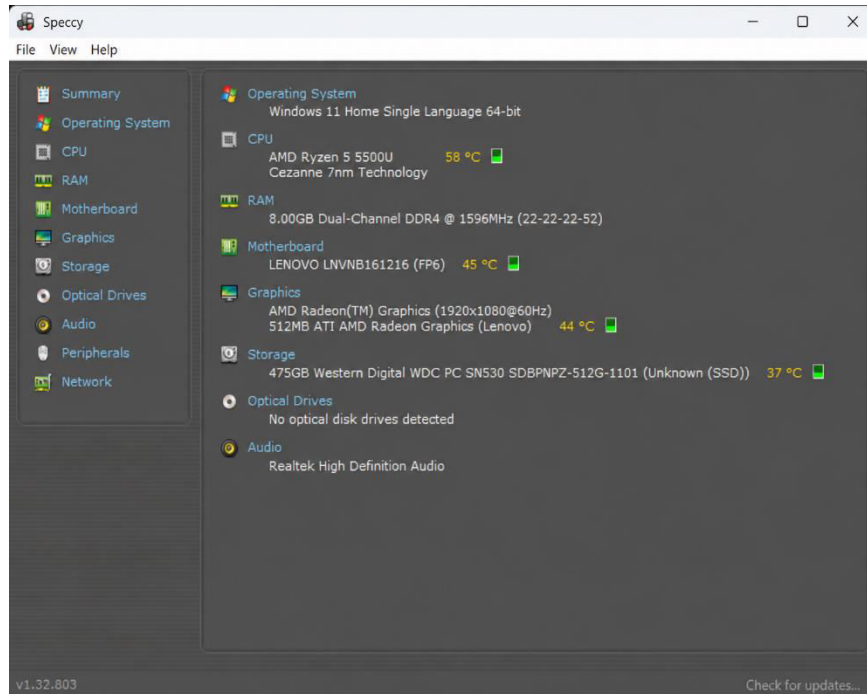
2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.



- The POST card displayed two-digit hexadecimal POST codes during the motherboard startup process.
- The display stopped at code **55**, indicating a RAM initialization problem.
- The technician checked and reseated the RAM modules to troubleshoot the issue.
- After fixing the RAM problem, the system booted normally, showing how the POST code helped identify the faulty hardware.

- Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.
Constraint: Blur or crop out any personal information before submitting.



- Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.



Tool	What it Checks	Ex Problem Detected
Multimeter	Electrical values such as voltage, current, and continuity	Faulty laptop charger not providing correct voltage
POST Card	Motherboard POST (Power-On Self-Test) error codes during startup	RAM failure causing the computer to stop booting
Software Diagnostics	System health, hardware status, temperatures, and disk information	Hard drive health warning detected by CrystalDiskInfo